Class 2 Mathematics

Expanded Practice Workbook

100+ Advanced Problems with Working Space

Chapter 1: Numbers up to 1000

Section A: Core Number Concepts

1. Write the number name for **345**.

2. Write the number name for **802**.

3. Write in numerals: Four hundred fifty-six.

4. Write in numerals: Nine hundred nine.

5. What number comes just before **500**?

6. Write the short form: **400 + 20 + 7** = ☐

7. Circle the largest number: **345, 354, 435, 453**

8. Circle the smallest number: **810, 801, 108, 180**

9. Arrange in ascending order: **231, 132, 312, 213**

10. Make the largest 3-digit number using 4, 1, 9.

Section B: Application of Numbers

11. Atharv is organizing his toy blocks. He has 3 boxes of 100, 4 stacks of 10, and 6 loose blocks. What is the total number of blocks he has?

12. Anaya is looking at a picture book. She is on page 145. She turns 10 more pages. What page is she on now?

13. In a factory producing Crunchley snacks, the machine packed batch number 899. What will be the number of the next batch?

Chapter 2: Addition (2 and 3 Digits)

Section A: Addition Practice Drill

1. 45 + 32 = ☐

2. 61 + 28 = ☐

3. 142 + 235 = ☐

4. 304 + 192 = ☐

5. Add: 24 + 13 + 31 = ☐

6. Find the sum of 365 and 244.

7. Fill the missing digit: 4_ + 23 = 68

8. What is 10 more than 485?

9. ____ = 300 + 450

10. Evaluate: 99 + 99 + 2

Section B: Addition Word Problems

11. While visiting a cherry orchard in California, Priyanka picked 145 cherries and Atharv picked 128 cherries. How many cherries did they pick in total?

12. For a family party, a restaurant prepared 150 idlis and 120 medu vadas. How many items were prepared altogether?

13. A library in Milpitas has 340 children's books on the first shelf and 285 on the second shelf. What is the total number of children's books?

Part 2/3/4: Multi-Concept Combined Sets

Level 1: Core Proficiency (Average Student) - SET 1

1. Add 145 and 32, then subtract 10 from the result.

2. Arrange in descending order: 450, 405, 540, 504.

3. There are 5 plates. Each plate has 3 idlis. How many idlis are there in total?

4. Write the expanded form of 749.

5. Atharv had 100 rupees. He bought a notebook for 45 rupees. How much money does he have left?

6. What number comes exactly between 299 and 301?

7. A box holds 10 packets of Crunchley makhana. How many packets are in 8 boxes?

8. Skip count by 5s: 15, 20, 25, ___, ___, ___

9. Add the numbers: 200 + 40 + 6.

10. Anaya picked 24 flowers and her friend picked 31 flowers. How many flowers did they pick together?

Part 2/3/4: Multi-Concept Combined Sets

Level 1: Core Proficiency (Average Student) - SET 2

1. What is the place value of 6 in the number 862?

2. Solve: 300 + 45 - 20 = ?

3. Priyanka baked 40 cookies. She gave 12 to Atharv. How many cookies are left?

4. Write the number name for 615.

5. If a bicycle has 2 wheels, how many wheels do 7 bicycles have?

6. Which is smaller: 809 or 890?

7. Find the sum of 150 and 225.

8. Subtract 45 from 100.

9. A tree has 85 apples. 20 apples fall to the ground. How many apples are still on the tree?

10. Count backwards by 10s: 90, 80, 70, ___, ___, ___

Part 2/3/4: Multi-Concept Combined Sets

Level 2: Advanced Application (Top 10%) - SET 1

1. Fill in the missing digits: 4_2 + _35 = 687

2. Priyanka packed 120 boxes of snacks on Monday and 155 on Tuesday. On Wednesday, she shipped out 200 boxes. How many boxes are still left in the warehouse?

3. If 4 apples cost 20 rupees, what is the cost of 1 apple?

4. I am an even number. I am greater than 140 but less than 144. Who am I?

5. Complete the pattern: 10, 20, 35, 55, ___ (Hint: Look at the difference between numbers).

6. Atharv reads 15 pages of his book every day. How many pages will he read in 4 days?

7. Add the largest 2-digit number to the smallest 3-digit number.

8. There are 350 books in the Milpitas library children's section. 125 are checked out, and 40 are new arrivals waiting to be shelved. How many books are currently on the shelves?

9. Find the missing number: ___ - 145 = 355

10. Look at the numbers: 12, 15, 18, 21. What is the rule for this pattern?

Part 2/3/4: Multi-Concept Combined Sets

Level 2: Advanced Application (Top 10%) - SET 2

1. Subtract the smallest 2-digit number from the largest 2-digit number, then add 5.

2. Atharv reads 3 books a month. How many books will he read in an entire year (12 months)?

3. A train has 450 seats. 234 people board the train at the first station. 102 more board at the second station. How many seats are still empty?

4. I have 3 hundreds, 14 tens, and 5 ones. What number am I?

5. A carton holds 6 bottles of juice. How many cartons are needed to pack 48 bottles?

6. What is the difference between the face value and the place value of 7 in the number 472?

7. Anaya has 45 stickers. She gives 15 to her friend, and then buys 20 more. How many stickers does she have now?

8. Fill in the blanks: 5 x 4 = 10 x ___

9. A shopkeeper had 500 packets of makhana. He sold 215 on Monday and 180 on Tuesday. How many packets are left?

10. Which two numbers add up to 150? (A) 100 and 40 (B) 80 and 70 (C) 90 and 50

Part 2/3/4: Multi-Concept Combined Sets

Level 3: Logical Reasoning (Top 1%) - SET 1

1. A ribbon is 40 cm long. You cut it into pieces that are each 8 cm long. How many cuts do you need to make? (Careful!)

2. If today is Tuesday, what day of the week will it be 15 days from now?

3. I have coins of 10 rupees and 5 rupees. I have a total of 35 rupees using exactly 4 coins. How many of each coin do I have?

4. The sum of two numbers is 20. Their difference is 4. What are the two numbers?

5. Atharv wrote numbers from 1 to 20. How many times did he write the digit '1'?

6. A clock shows the time as 3:00. The minute hand moves 30 minutes ahead. What time does the clock show now?

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7. In a farm, there are cows and chickens. There are 5 heads and 16 legs in total. How many cows and how many chickens are there?

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8. Find the missing number in the sequence: 2, 4, 8, 14, 22, ___

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9. Anaya is taller than Rahul but shorter than Priya. Amit is the tallest. Who is the shortest among the four?

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10. You have a 5-liter jug and a 3-liter jug. How can you measure exactly 2 liters of water?

Part 2/3/4: Multi-Concept Combined Sets

Level 3: Logical Reasoning (Top 1%) - SET 2

1. If you multiply me by 3 and add 5, you get 26. What number am I?

2. A pizza is cut into 8 equal slices. Atharv eats half of the pizza, and Anaya eats 1 slice. How many slices are left?

3. There are 5 people in a room. If everyone shakes hands with everyone else exactly once, how many handshakes happen in total?

4. A book has pages numbered from 1 to 50. How many times does the digit '4' appear in all the page numbers combined?

5. I am a 3-digit number. My tens digit is 5 more than my ones digit. My hundreds digit is 8 less than my tens digit. What number am I?

6. If 3 cats can catch 3 mice in 3 minutes, how many cats are needed to catch 100 mice in 100 minutes?

7. Look at this pattern: A, C, E, G. What is the next letter?

8. Priyanka plants seeds in her garden in a row. The 1st seed is a tomato, 2nd is a pepper, 3rd is a tomato, 4th is a pepper. What will the 11th seed be?

9. How many 2-digit numbers can you make using the digits 1, 2, and 3 without repeating any digit in the same number?

10. A log of wood takes 10 minutes to cut into 2 pieces. How long will it take to cut it into 4 pieces?

Part 2/3/4: Multi-Concept Combined Sets

Level 4: Critical Thinking (Top 0.1%) - SET 1

1. A snail is climbing up a 10-foot wall. During the day, it climbs up 3 feet. At night, while sleeping, it slips down 2 feet. How many days will it take for the snail to reach the top of the wall?

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2. If 🔺 + 🔺 + 🔺 = 15, and 🔺 + 🟦 + 🟦 = 13. What is the value of 🔺 x 🟦?

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3. Atharv has a secret 3-digit code. The hundreds digit is half of 8. The tens digit is the hundreds digit minus 1. The ones digit is the sum of the hundreds and tens digits. What is the code?

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4. How many squares can you find on a normal checkerboard (8x8) if you only count 1x1 squares and 2x2 squares? (Hint: Think logically!)

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5. Three friends—A, B, and C—are sitting in a row. A is not on the right end. B is to the right of A. C is to the left of A. What is their seating order from left to right?

6. I have a certain number of candies. If I divide them equally among 3 friends, I have 1 left over. If I divide them equally among 4 friends, I have 1 left over. What is the smallest number of candies I could have?

7. Fill in the magic square (3x3 grid where all rows, columns, and diagonals sum to 15) using numbers 1 to 9. The center number is 5. What are the corner numbers?

8. A water tank is half full. If you add 10 liters, it becomes three-quarters full. What is the total capacity of the tank?

9. What is the next number in the sequence: 1, 1, 2, 3, 5, 8, ___?

10. You have 9 marbles. One is slightly heavier than the rest. Using a balance scale, what is the minimum number of weighings needed to find the heavy marble?

Part 2/3/4: Multi-Concept Combined Sets

Level 4: Critical Thinking (Top 0.1%) - SET 2

1. If 2 typographical errors are found on every page of a 10-page book, and you fix half of them, how many errors remain?

2. A doctor gives you 3 pills and tells you to take one every half hour. How long will the pills last?

3. Some months have 31 days. How many have 28?

4. If you have a drawer with 10 black socks and 10 white socks, and you close your eyes, how many socks must you pull out to guarantee a matching pair?

5. I am an odd number. Take away one letter and I become even. What number am I?

6. A father and son have a combined age of 66. The father's age is the son's age reversed. How old could they be? (Give one possible answer).

7. Using exactly four 4s and any math operations (+, -, x, /), make the number 5.

8. A bat and a ball cost $1.10 in total. The bat costs $1.00 more than the ball. How much does the ball cost?

9. If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets?

10. There is a patch of lily pads on a lake. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how long would it take for the patch to cover half of the lake?

Part 2/3/4: Multi-Concept Combined Sets

Level 5: National Olympiad (Top 10) - SET 1

1. In a race, Atharv is the 15th fastest runner and also the 15th slowest runner. How many runners are participating in the race?

2. You have a pan that can hold 2 dosas at a time. It takes 1 minute to cook one side of a dosa. How long will it take to cook 3 dosas on both sides in the shortest possible time? (Think outside the box!)

3. How many 2-digit numbers can you make using the digits 0, 1, and 2 if you are allowed to repeat digits? (Remember, a 2-digit number cannot start with 0).

4. There are 5 locked chests and 5 keys. Each key opens exactly one chest. In the worst-case scenario, how many attempts (inserting a key into a lock) must you make to guarantee you match all keys to their correct chests?

5. A wooden cube is painted completely red on the outside. Then, it is cut into 27 smaller, equal-sized cubes (like a 3x3x3 Rubik's cube). How many of the smaller cubes have NO red paint on them at all?

6. In a strange language, 'Maka' means 'Red', 'Baka' means 'Car', and 'Maka Baka Shaka' means 'Fast Red Car'. What does 'Shaka' mean?

7. How many rectangles are there in a 2x2 grid? (Count all sizes: 1x1, 1x2, 2x1, and 2x2).

8. You have bags of coins. One bag has fake coins that weigh 9 grams each, while real coins weigh 10 grams. You have a digital scale. How can you find the fake bag in just ONE weighing?

9. A frog is at the bottom of a 30-meter well. Each day he leaps up 3 meters, but slips back 2 meters at night. On which day does the frog finally escape the well?

10. What is the maximum number of pieces you can cut a circular cake into with just 3 straight cuts?

Part 2/3/4: Multi-Concept Combined Sets

Level 5: National Olympiad (Top 10) - SET 2

1. If a train is traveling at 60 km/h, how far will it travel in 45 minutes?

2. You are given two ropes. Each rope takes exactly 1 hour to burn completely, but they burn at inconsistent rates. How can you measure exactly 45 minutes?

3. In a tournament with 16 tennis players, it is a knockout format (single elimination). How many matches must be played to determine the winner?

4. You have 10 stacks of 10 coins each. One stack contains fake coins that are 1 gram lighter than the real coins. Using a digital scale only once, how do you find the fake stack?

5. If A + B = 10, B + C = 15, and A + C = 11, what is the value of A + B + C?

6. A grandfather, two fathers, and two sons went to a baseball game together and bought one ticket each. How many tickets did they buy in total?

7. Find the missing number in the grid logic puzzle: Top row (3, 6, 9), Middle row (2, 4, 6), Bottom row (5, 10, ?)

8. What number comes next? 1, 11, 21, 1211, 111221, ___ (Hint: Read it out loud)

9. You have 100 doors in a row, all closed. You make 100 passes. On pass 1, you toggle every door. On pass 2, you toggle every 2nd door. On pass 3, every 3rd door, and so on. Which doors are open after 100 passes?

10. If you paint the outside of a 4x4x4 cube and break it into 1x1x1 cubes, how many of the small cubes have exactly 2 painted faces?